

AssessClear Diagnostic Intelligence: Pre-Pilot Evaluation & Strategic Action Plan

Comprehensive Field Findings, Usability Assessment, Gap Analysis, Root Causes & Prioritized Product Roadmap

EVALUATION LEAD

Senior Evaluation Specialist

PRE-PILOT SCOPE

9 Schools (Katni, Indore, Bhopal, Raisen)

EVALUATED SAMPLE

~351 Student Copies · Grades 6–8

MODEL ENGINE

Qwen 3 (14B GPU) + Vision Core

1. Executive Summary & Readiness Verdict

Pre-Pilot Readiness Verdict: **REQUIRES ATTENTION BEFORE PILOT (Conditional Green Light)**

The core diagnostic value proposition of automated multi-tiered feedback (1.0, 0.75, 0.40, 0.0) is proven and validated in field classrooms. However, critical software/vision defects (Q1/Q2 question skipping, digit misreading, rigid semantic matching) and operational bottlenecks ("double effort" workflow taking 80–90 min per 50 sheets) must be addressed before wide deployment.

Validated Value Propositions

- **Constructive Diagnostic Feedback:**
Hand-holds students with 3-part remedial steps (*What you did well, Think about this, Next steps*).
- **Partial Attempt Credit:**
Rewards partial understanding and correct mathematical approach (0.40 / 0.75 weights).
- **Hindi OCR Relevance:**
Directly supports Hindi-medium government classrooms across MP.

Key Pilot Vulnerabilities

- **Question Dropping:**
Vision pre-processing clips top/bottom questions (10 attempted → 7 evaluated).
- **Teacher Overhead:**
1–3 min review per paper creates a high administrative burden for classes of 30–50+ students.
- **Zero-Network Failure:**
Rural schools (MS Kharbai) freeze completely without offline sync.

Table 1: Pre-Pilot Findings & District Evidence Matrix

District & Schools	Sample Evaluated	Key Field Strengths	Critical Observed Defects
Indore (Nandbag & Subhash Nagar)	100 Students (60 Math, 40 Hindi) DIET Observer	• Rapid batch upload efficient (≤ 10 sheets). • 20–30s AI processing time. • Constructive feedback validated by DIET.	• Q1 and Q2 skipped on first page. • Rigid logic rejected "3 months" when key was "90 days". • Incorrect arithmetic box sum marked correct. • Dashboard omitted Hindi and Grade 7–8 data.
Bhopal & Raisen (5 Schools incl. MS Kharbai)	~160 Students Grades 6–8 (Math & Hindi)	• In-app direct scanning saves local phone memory. • 3-part structured feedback easy to understand. • Hindi language relevance confirmed.	• Zero network in MS Kharbai halted all scanning. • Reviewing 30+ students individual feedback took >80 min. • Downloading individual PDFs fills phone storage. • Feedback focused on "what" rather than "how to do it".
Katni (EPES Purwar & MS Devri Hatai)	91 Students (32 Purwar, 59 Hatai) Grade 6 Math/Hindi	• Step-by-step remedial handholding praised. • Encourages learning beyond binary right/wrong.	• Dashboard showed 29 evaluations for 3 copies. • Session codes hard to retrieve without manual notes. • Digit misrecognition (9996 → 3226). • Corrected pencil answers ignored; 200-eval cap drops data.

Table 2: User Experience (UX) & Usability Evaluation Matrix

UX Dimension	Positive Design Elements	Observed Field Friction & Usability Gaps
Navigation & Access	Clean, intuitive role-based views for DLs, School Admins, and Teachers.	Dashboard filtering clutter; missing subject tabs; inconsistent grade views.
Session Management	Amber session codes allow resuming work across different sittings.	Codes are not saved in user profile; if not jotted down manually, sessions become orphaned.
Capture & Ingestion	Rapid batch upload simplifies scanning up to 10 sheets simultaneously.	One-by-one mode has page omission issues; lack of offline queue freezes app in low network.
Review & Action	Rich misconception tags provide deep diagnostic insight.	"Double Effort" dilemma: Teachers spend 3–4 min/paper transcribing remarks onto physical sheets.
Language & Localization	Direct ingestion of handwritten Hindi scripts.	Hindi PDF output contains awkward grammar and phrasing; student feedback missing in PDF.

Table 3: Gap Analysis Matrix (Expected Design vs Observed Reality)

System Dimension	Designed / Intended Specification	Actual Pre-Pilot Experience	Districts Affected
Question Detection	AI reads and grades 100% of questions from sheet.	Skipped top/bottom questions (Q1, Q2); only 7 of 10 scored.	Indore, Katni
Semantic Grading	Flexible diagnostic evaluation against learning outcome.	Rigid exact matching: marks 3 months wrong vs 90 days .	Indore
Offline Resilience	PWA handles rural 3G/4G connectivity smoothly.	Zero offline support; complete failure in zero-network areas.	Raisen (MS Kharbai)
Dashboard Retention	Continuous cohort SLO trend tracking.	Counts questions rather than copies; caps at ~200 items.	Katni, Indore
PDF Export	Dual Teacher Instructions + Student Feedback PDF.	Exported PDF shows only teacher feedback; student view lost.	Katni
Teacher Autonomy	Teacher can review and calibrate AI assessments.	No in-app 1-click override or score correction button.	Indore, Bhopal

Table 4: Issue, Root-Cause & Severity Analysis Table

Identified Issue	Direct Evidence Base	Technical / Pedagogical Root Cause	Severity
Question Segmentation Defect	Q1 & Q2 skipped in Indore & Katni	Auto-crop bounding box margins clip top/bottom headers of worksheet image.	P0 (CRITICAL)
Handwriting & Digit Errors	9996 → 3226 , marked D read as B	Insufficient prompt calibration and vision pre-processing for pencil handwriting.	P0 (CRITICAL)
Evaluation Count & Cap Defect	29 evals for 3 copies; history drops >200	Firestore schema tracks question IDs instead of student submissions; missing query pagination.	P0 (CRITICAL)
Rural Network Dependency	MS Kharbai scanning lockout (Raisen)	Absence of Service Worker offline caching and IndexedDB background sync queue.	P1 (HIGH)
Session Code Friction	Evaluators unable to resume sessions	Active sessions not persisted in user profile or visible in dashboard dropdown.	P1 (HIGH)
High Review Time Overhead	3–4 min/paper review (80–90 min/class)	Absence of aggregated Class-Level Summary Heatmaps; forces 1-by-1 screen review.	P1 (HIGH)

Table 5: Pilot Readiness Assessment & Strategic Risk Register

Readiness Status	Key Subsystem Diagnostics	Mandatory Pre-Pilot Mitigation Strategies
P0 BLOCKER	AI Question Segmentation: High risk of truncating valid student test responses.	Adjust crop padding, expand vision bounding boxes, and enforce schema count verification.
P0 BLOCKER	Data Schema & Retention: 200-evaluation cap truncates historical reporting.	Refactor database documents to track student submissions; paginate dashboard history.
P1 HIGH RISK	Teacher Adoption Risk: High time burden leads to rejection during pilot.	Deploy Class-Level Summary UI to reduce total review time per batch to <5 minutes.
P1 HIGH RISK	Connectivity Risk: Field teams locked out during rural visits.	Implement offline scanning with automatic background sync upon network availability.

Table 6: Strategic Action Roadmap & Prioritization Table (P0, P1, P2, P3)

Pri.	Strategic Recommendation	Operational Rationale	Timeline
P0	Fix Vision Segmentation & Blur Gate	Eliminates Q1/Q2 skipping and prevents AI hallucinations on blurry images.	Immediate (Week 1)
P0	Refactor Dashboard Schema & Pagination	Corrects evaluation counts (copies vs questions) and lifts the 200-eval limitation.	Immediate (Week 1)
P0	Implement Offline-First PWA Caching	Enables uninterrupted scanning and local storage in zero-network rural schools.	Immediate (Week 2)
P1	Build Class-Level Summary Dashboard	Reduces teacher review overhead from 80–90 min to <5 min for a class of 40 students.	Immediate (Week 2)
P1	Add Teacher Override / Edit Button	Empowers educators to correct AI scores, protecting trust and data integrity.	Immediate (Week 2)
P1	Active Session Drawer in Profile	Eliminates lost session codes; allows 1-click session restoration.	Immediate (Week 2)
P2	Refine Hindi Prompts & PDF Exports	Ensures high-quality Hindi grammar and formats dual Teacher/Student PDF exports.	Pre-Pilot Polish
P2	Deep Integration with Sakhi AI	Triggers automated WhatsApp lesson plans aligned to red-band class competencies.	During Pilot

Table 7: Pilot Monitoring, Metrics & Evaluation Framework

Evaluation Category	Key Performance Metric	Pilot Target Benchmark	Verification Method
Diagnostic Accuracy	Question Detection Rate AI Scoring Agreement Rate	> 99.0% detection > 95.0% scoring agreement	Compare AI scores against blind expert teacher evaluations.
Operational Speed	AI Processing Latency Teacher Review Overhead	< 25 sec per sheet < 15 sec per student	In-app telemetry logs and field time-motion observations.
Field Reliability	Offline Queue Sync Fidelity Session Restoration Rate	100% data sync > 98.0% session recovery	Track background sync logs across low-connectivity clusters.
Teacher Engagement	Weekly Active Usage Sakhi AI Remedial Follow-up	> 75% weekly adoption > 60% remedial resource use	Dashboard analytics and Sakhi AI WhatsApp interaction telemetry.

8. Final Analytical Conclusion

AssessClear represents a **transformative diagnostic innovation** for government school education, replacing delayed, high-overhead manual grading with real-time, multi-tiered misconception analysis. The pre-pilot exercise in Madhya Pradesh successfully proved the viability of mobile multimodal grading at ₹0.12/eval.

By executing the **P0 blocker resolutions** (crop boundary correction, image quality gate, offline sync, and class-level summary screens) before full rollout, project leadership can ensure a high-fidelity, teacher-friendly, and statistically robust pilot deployment across the state.